THE CHINESE ECONOMY

A JOURNEY INTO HISTORY AND ECONOMICS

Lecture 1: Foundations


Political
Physical


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Pudong skyline at night, Shanghai
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History

"…the science of men in time."

Marc Bloch

we will use it

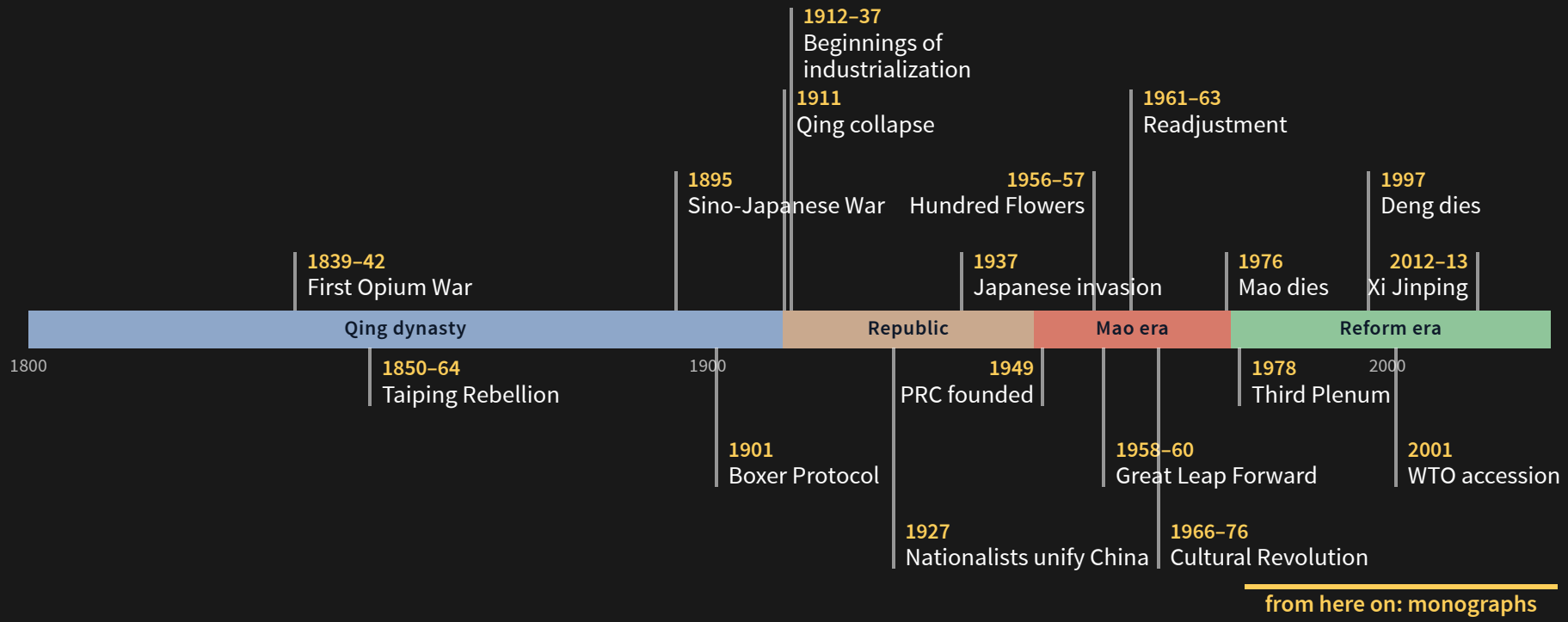
Economics

"…the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."

Lionel Robbins, 1932

we will do it

150 YEARS IN ONE LINE



THE COURSE IN SEVEN BLOCKS

- **Foundations** — lecture 1 · history and institutions, from 1800 to 1978 *(today)*
- **Causality** — lectures 2-3 · the method: what a cause is, and how economists find one
- **Reallocation across sectors** — lectures 4-5 · from the farm to the factory: GDP, hukou, migration
- **Reallocation within sectors** — lecture 6 · firms, state-owned enterprises, misallocation
- **Trade** — lectures 7-8 · growth, the WTO, tariffs and trade wars
- **Innovation** — lectures 9-11 · China at the Frontier
- **Conclusions** — lecture 12

RULES OF THE JOURNEY

- **Website** — syllabus and slides: **framana111.github.io/the-chinese-economy**
- **Exercises** — from time to time, a short problem to take home: tools to learn the logic of economics, analysis and reflection; **not graded**
- **Assessment** — final written exam

A QUICK POLL

Yardsticks

Average annual **real GDP growth, 2005-2024**

- **Italy** 0.3% a year
- **European Union** 1.3% a year
- **United Kingdom** 1.4% a year
- **United States** 2.1% a year
- **Japan** 0.6% a year

CHINA, 2005-2024

7.9% a year — real GDP, 20 years

- The economy became **4.6 times** larger (+356%)
- Italy +5% · European Union +30% · United Kingdom +30% · United States +51% · Japan +12%

POLL 2

What drove this growth? Rank all six, from 1 (most important) to 6 (least)

- movement of workers from the countryside to the cities
- reform of state enterprises and better allocation across firms
- foreign trade and WTO entry
- innovation and technology catch-up
- high investment and saving
- industrial policy and state direction


Aihui
West: 6% of the population
Tengchong
East: 94%

CHINA, AROUND 1800

- About 381 million people in 1820 — around **36% of the world**
- About a third of world GDP (32%, Maddison); living standards close to the world average
- Over 90% rural: very high yields per hectare, low output per worker
- A dense, commercial economy: markets, money, contracts, competition — small and household-based ("bottom-heavy")

. . .ALREADY UNDER STRAIN

- Land: nearly all cultivable land in use by 1800 — "high-level equilibrium trap" (Elvin 1973)
- State: a thin layer — revenue 2-3% of national income; one official per 32,000 people (Europe: one per 600-800)
- Inequality: the gentry (under 1% of people) took about 20% of income
- "The Chinese peasant is like a man standing on tiptoe up to his nose in water" (Tawney)



Johanna Helena Herolt, *A Poppy in Three Stages of Flowering, with a Caterpillar, Pupa and Butterfly*, late 17th-early 18th century.
Metropolitan Museum of Art (CC0)

TRADE, SILVER AND OPIUM





The opium fleet on the Ganges, c. 1850. Lithograph after W. S. Sherwill (Wellcome, CC BY 4.0)

- For centuries China sold tea, silk, porcelain and received **silver**
- "British merchants were unhappy with the steady drain of silver into China. […] They finally located such a commodity in **opium**." — *Naughton (2007), p. 41*
- By the 1830s China imported more than it exported

WHAT THE OPIUM SURGE DID



Two wealthy Chinese opium smokers. Gouache on rice paper, 19th century (Wellcome Collection, CC BY 4.0)

- Imports of Indian opium doubled in five years: 21,885 chests (1834/35) to 40,200 (1838/39) — *Greenberg, via Farooqui (2016)*
- More users, but no reliable count: estimates run from a few million to over ten million — *MIT Visualizing Cultures*
- It reached officials, soldiers and the court; local officials took bribes to ignore the ban — *MIT Visualizing Cultures*
- The court split between prohibition and legalisation — *Polachek (1992)*
- Silver outflow and currency stress: usually blamed on opium, a link recent work questions — *Lin (2006); Irigoin et al.*

THE OPIUM WARS

- **1839-42** — China tries to stop the opium inflow; Britain attacks. *Treaty of Nanjing* (1842): Hong Kong ceded, five treaty ports opened, indemnity of 21 million silver dollars, end of the Cohong monopoly
- **1856-60** — second war; treaties of Tianjin (1858) and Convention of Peking (1860): more ports, envoys in Beijing, missionaries, opium legalised
- Through 1895 four more wars, all lost: reparations, more treaty ports (over 80 at the peak)



The *Nemesis* destroying Chinese war junks, Anson's Bay, 7 January 1841. Print by Edward Duncan, 1843 (public domain)

FROM BRITAIN TO JAPAN

- **1895**: Sino-Japanese War → Taiwan to Japan; the Treaty of Shimonoseki lets foreigners invest — modern industry begins
- **1901**: after the Boxer Rebellion, eight powers occupy Beijing; Boxer Protocol: indemnity, tariff revenue under foreign control
- **1911**: the Qing dynasty collapses
- Share of world GDP: **32% (1820) → 9% (1913)**

1912-1937: FIRST INDUSTRIALIZATION

- After 1911: warlords; **1927** the Nationalists unify the country; the "Nanjing decade"
- Modern factories grow **8-9% a year** — from 2% of GDP and 0.4% of the labour force
- Two patterns: **Treaty-Port** industry (light, Chinese-owned by the 1930s) and **Manchuria** (Japanese, heavy, few spillovers)

1937-1949: WAR AND CIVIL WAR

- **1931** Manchukuo; **1937** Japan invades China proper — a decade of war
- Manchuria's industry grows 14% a year (1936-42); Shanghai never regains its 1936 output before 1949
- The Nationalist state runs the war economy; **hyperinflation**: prices in 1948 = 660 million (1936 Shanghai = 100)
- 1949: the Communist victory

WHERE THIS LECTURE GOES

- **1800 → 1949** — how a great economy arrived at the revolution
- **1949 → 1978** — a planned economy: how it worked, what it did, and the Great Leap Forward
- **1978** — the turn, and a question we cannot yet answer

THE ECONOMIC PROBLEM

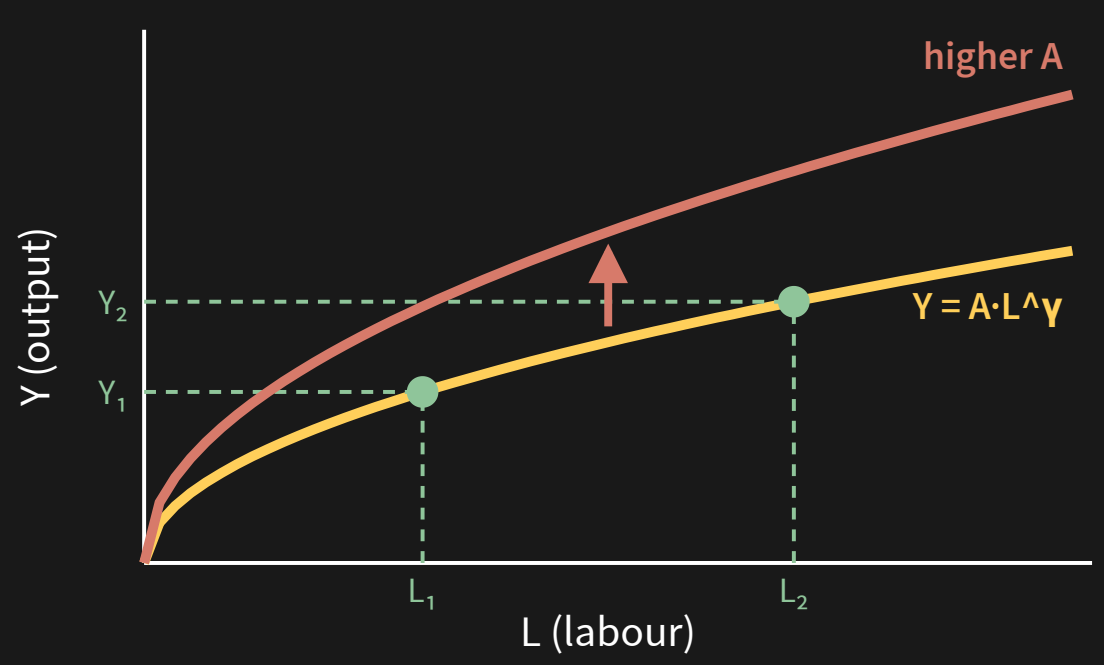
Who gives what to whom — and who decides?

- Millions of different actors: households, farms, factories
- Each must decide how much to take and how much to give to the others
- Two ways to coordinate: through **prices**, or through **quantities** (a plan)

WHAT DOES A PRICE DO?

- It says how **scarce** something is, relative to everything else
- It carries information nobody holds in full — *Hayek (1945)*: knowledge is dispersed; prices coordinate separate actions
- It rewards whoever economises or finds a substitute
- *Other traditions: prices also reflect technology and the distribution of income (Ricardo, Marx, Sraffa)*

HOW PRODUCTION WORKS



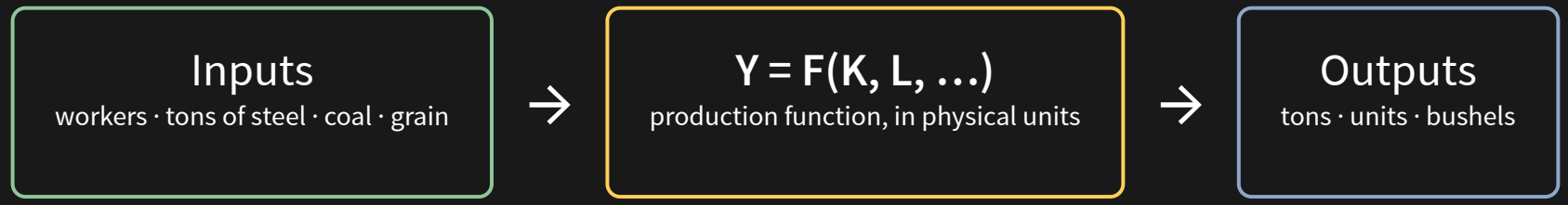
Y = A · L$^{\gamma}$, with 0 < γ < 1

Y output · **L** labour, the only input for now · **A** productivity

More L: we move **along** the curve, to a higher Y. Each extra worker adds less: diminishing returns

Higher A: the **whole curve** shifts up. Same labour, more output

A PLAN WORKS ON QUANTITIES



- The planner sets **targets** and allocates **quantities** to each producer
- Prices are still there — but set by the state, not by the market

CHINA'S COMMAND ECONOMY

- The state owns the big factories; collectives own the land
- **Commands**: production targets; resources allocated directly — prices lose their role as the main signal
- The state still *sets* prices — industry dear, agriculture cheap — to channel resources to itself; compulsory grain procurement from 1953
- The Party controls managers' careers
- Labour fixed in place: tight limits on movement, jobs for life
- Scale: Beijing allocated at most 600 product varieties; the USSR, 60,000 by the 1970s

MEASURING WITHOUT MARKET PRICES

- GDP adds up value at market prices — what is "value added" without them?
- China used the **Material Product System (MPS)** until 1992: it counts material production (agriculture, industry, construction, freight transport, distribution) but not *non-material* services (education, health, finance, housing, administration)
- MPS and GDP coexisted in 1985-92; GDP only from 1993
- So we look at a different indicator, the planners' main lever: **investment**

THE PLANNERS' LEVER: INVESTMENT

- Gross capital formation was already **26% of GDP in 1954**; it peaked near **43% in 1958-59**, fell to about **15% in 1962**, then rose again to 30-35%
- A year in which investment grows by more than **20%** is a "leap": 1953, 1956, 1958-59, 1963-66, 1970, 1978
- Between the leaps, retrenchments: 1960-62, 1967

Source: Naughton (2007), The Chinese Economy, Figs. 3.1 and 3.2 — values read from the charts, approximate

LEAPS AND RETRENCHMENTS

- A **leap**: investment grows more than 20% in a year — rapid growth *and* political mobilisation; radical, "leftist" phases
- Then **retrenchment**: slower investment, consolidation, often a more moderate climate (Hundred Flowers 1956-57; readjustment 1961-63; drift 1972-76)
- "Policy swings like a pendulum from left to right and back again" — and drives, and is driven by, economic developments

1958 — HUKOU

- **1955**: grain rationing begins (lasts over 30 years)
- **1958**: household registration formally instituted — working outside one's hukou place or category is prohibited
- After the Leap: urban registration a "meal ticket", rural registration "a license to starve"
- 1961-62: six million recent urban arrivals sent back to the villages; by the mid-1960s, a "caste system"

THE GREAT LEAP FORWARD, 1958-60

- "A vision, rather than a plan": do everything at once, regardless of real resource constraints
- Communes; bonuses and free markets abolished; decisions decentralised; "walking on two legs"
- Two fateful decisions: **less labour and land for farming; more grain taken by the state**
- Inflated reports: the statistical system, the leaders' "eyes and ears", collapsed

THE PUZZLE

In 1959-61, average food availability in the Chinese countryside was enough to feed the population.

So why did 16.5 to 45 million people die?

THE INSTITUTIONAL MECHANISM

Meng, Qian, Yared (2015), *Review of Economic Studies* 82(4), 1568-1611

- Excess deaths: **16.5-45 million** (an explicit range; Naughton: 25-30 million through 1961)
- Per capita grain output fell about 15% in 1959 — but stayed above subsistence
- Regions with **higher** production had **higher** mortality
- Explanation: **fixed quantity procurement** and a plan that cannot collect or respond to information after an aggregate shock

HOW DO WE KNOW? DATA, PROXIES, CLAIMS

- There is no county-level data on output or deaths for the famine
- Proxies: missing birth cohorts, land suitability for grain, official (inflated) production figures
- The paper **documents** facts and **argues** for an institutional story
- What it does not yet tell us: what would have happened otherwise

1961-76 — READJUSTMENT, THIRD FRONT, CULTURAL REVOLUTION

- **1961-63** readjustment: investment chopped back, about 20 million workers sent to the countryside
- **1964-66** Third Front: a new expansion into the interior, "hijacked by radicalism"
- **1966-76** Cultural Revolution: 17 million urban school-leavers sent to the countryside (*xiafang*); schools and universities closed
- *Seed*: small rural industry — the precursors of the TVEs (lecture 6)
- **1976**: Mao dies

1976-78 — THE TURN

- Growth had accelerated under the Big Push — but even more in Taiwan and Hong Kong, far ahead of the PRC by 1978
- That demonstration effect convinced Chinese leaders to reform and open up
- **December 1978**: Third Plenum — ease the terms of trade with agriculture, "give farmers a chance to catch their breath"
- Deng Xiaoping consolidates power; *gaige kaifang*, reform and opening

A QUESTION

*China grew about **10% a year** from 1978 to 2005 (official data).*

Did the reforms cause it — or would China have grown fast anyway?

- Economists: continuity — a traditional economy well suited to growth
- Historians: discontinuity — 1949 was a social revolution
- Hong Kong and Taiwan accelerated even more, without socialism

Think about it. → Lecture 2: how do we answer a question like this?